

Ilan Sela

Texas A&M University
Dept. of Electrical & Computer Engineering

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RESEARCH INTERESTS

Generative modeling and theoretical machine learning.

EDUCATION

Texas A&M University

M.S. in Computer Engineering
Advisor: Dr. Tie Liu. *GPA: 4.0/4.0*

College Station, TX
Expected: May 2027

Texas A&M University

B.S. with honors in Electrical Engineering
Minor in Computer Science
GPA: 4.0/4.0

College Station, TX
2022 – 2025

RESEARCH EXPERIENCE

Texas A&M University

Graduate Researcher (with Prof. Tie Liu)

- Studying drift-based generative training dynamics to inform theoretical analysis of one-step image generators.

College Station, TX
May 2025 – Present

Graduate Researcher (with Prof. Dileep Kalathil)

- Developed AgentDiscover, an agentic framework for autonomous scientific discovery with minimal search scaffolding.
- Designed a persistent graph-based memory architecture enabling agents to retain and reuse discovered strategies across otherwise independent search iterations.

May 2026 – September 2026

Research Project

Reinforcement Learning-Based Calibration for Radar Systems

- Developed an offline RL framework for MIMO radar calibration without specialized measurement equipment.
- Formulated radar calibration as a Markov decision process and designed a reward function to maximize image quality.

January 2026 – May 2026

Undergraduate Researcher (with Prof. Tie Liu)

- Investigated LLM fine-tuning strategies for black-box offline optimization to generate highly optimized designs.
- Built custom training pipelines to optimize Google FLAN-T5 for the optimization task.

August 2024 – May 2025

Undergraduate Researcher

Secure and Trustworthy Hardware (SETH) Lab

- Implemented and tested a novel attack on embedded field-programmable gate arrays (eFPGAs), leveraging reinforcement learning (RL) to optimize attack efficiency.

May 2023 – May 2024

PUBLICATIONS AND RESEARCH OUTPUT

- Mahdi Farahbakhsh, **Ilan Sela**, Fatemeh Doudi, Vishnu Teja Kunde, Krishna Narayanan, Jean-Francois Chamberland, Dileep Kalathil. "AgentDiscover: Autonomous Discovery with Minimal Search Scaffolding." *arXiv:2610.05334*, 2026. *Under review*.
- Sumin Kim, **Ilan Sela**, Muhammad Faeyz Karim (2026). A Reinforcement Learning-Based Self-Calibration Framework for Radar Systems in Offshore Crane Applications. Presentation, 2026 Mary Kay O'Connor Safety & Risk Conference
- Ilan Sela** (2025). Black-Box Offline Optimization via Large Language Models. Undergraduate Research Scholars Program. Available electronically from <https://hdl.handle.net/1969.1/1591266>

ACADEMIC HONORS

Graduate

- Qualcomm Graduate Scholarship Recipient (2025 – 2026)

Undergraduate

- Undergraduate Research Scholar, Engineering Honors, Summa Cum Laude, Dean's Honor Roll (2022 – 2025)

TEACHING EXPERIENCE

Agentic Intelligence for Discovery and Exploration (AIDE) Workshop mentor, Texas A&M

August 2026

- Mentored faculty and graduate students in designing and implementing agentic workflows for research problems.

Student Assistant for "ECEN 740: Machine Learning Engineering", Texas A&M

January 2025 – May 2025

INDUSTRY EXPERIENCE

Luminator Technology Group Plano, TX

Embedded Software and Electrical Engineering Intern

May 2025 – August 2025

- Developed embedded firmware for microcontrollers used in Flying Whales aircraft systems.
- Developed CRJ cabin lighting retrofit system featuring wireless LED control, enabling rewiring-free installation that became the company's primary R&D focus.
- Engineered PCB designs in Altium: LED drivers and CAN bus distribution for aerospace lighting systems.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, MATLAB, Verilog, Java, Assembly

ML/Scientific Computing: PyTorch, TensorFlow, NumPy, SciPy

Embedded/Hardware: STM32, PIC, Altium

SELECTED COURSEWORK

- Machine Learning, Reinforcement Learning, Online Learning, Information Theory, Estimation and Detection Theory, Convex Optimization, GPU Programming